A Generated Split-Forest Decidability Proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$ Without Automata: A Repaired Containment-Oriented Presentation

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Abstract

We give a self-contained repaired proof of decidability for concept satisfiability in $\mathcal{ALCI}_{\mathrm{RCC5}}$ over strong abstract atomic RCC5 frames. The proof keeps the original split-forest architecture: a containment-oriented generated backbone for PP/PPI, split copies for multiple containment parents, finite sibling interfaces, and a typed equality quotient. The proof removes three sources of unsoundness. First, the vertical cover relation is a generated witness-cover relation, not the Hasse diagram of an arbitrary semantic PP-order. Second, blocking repetition is separated from semantic equality. Third, upward PP eventualities are discharged through equality-mate occurrences, so an element may have several incomparable selected superpart witnesses. No Buchi or parity automata are used. Regularity is obtained by a finite-index blocking lemma with explicit request-closed cycles, and the final decision procedure enumerates finite split-forest certificates and checks only finite local, reachability, equality-congruence, and mosaic conditions.

Contents

1 Statement of the theorem

The input is a concept C_0 in negation normal form over concept names and the five RCC5 atoms used as modal roles. The role set is

$$\mathsf{B} = \{\mathsf{EQ}, \mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}.$$

The intended result is the following.

Theorem 1.1 (Main decidability theorem)**.** *Concept satisfiability for $\mathcal{ALCI}_{\mathrm{RCC5}}$ over strong abstract complete atomic RCC5 frames is decidable. Equivalently, for every input concept C_0,*

$$C_0 \textit{ is satisfiable} \iff C_0 \textit{ has a valid finite generated split-forest certificate.}$$

The right-hand side is effectively enumerable and decidable by finite checks.

The proof has the following shape. First, every arbitrary satisfying abstract model can be thinned to a witness-generated induced submodel. This justifies working only with sparse generated witness-cover structures. Second, every witness-generated model has a split occurrence presentation in which vertical generated cover edges carry containment and multiple selected containment parents are represented by weak equality copies. Third, the possibly infinite split presentation can be replaced by a finite regular certificate by finite-index blocking. Blocking is not semantic equality:

different laps of a blocking cycle unfold to fresh occurrences. Finally, a finite validity test checks local types, exact PP/PPI reachability through equality mates, support-closed sibling mosaics, and typed equality congruence. Soundness unfolds a valid certificate to a strong abstract RCC5 model; completeness extracts a valid certificate from a satisfying model.

2 Abstract RCC5 frames

2.1 The relation algebra used in the proof

The inverse operation fixes $\mathsf{EQ}, \mathsf{DR}, \mathsf{PO}$ and swaps PP with PPI. The RCC5 composition table used below is the following. The entry in row R and column S is the set $\mathsf{comp}(R, S)$ such that, whenever xRy and ySz, the label of (x, z) must lie in $\mathsf{comp}(R, S)$.

$\mathsf{comp}(R,S)$	$S = \mathsf{EQ}$	$S = \mathsf{DR}$	$S = \mathsf{PO}$	$S = \mathsf{PP}$	$S = \mathsf{PPI}$
$R = \mathsf{EQ}$	$\{\mathsf{EQ}\}$	$\{\mathsf{DR}\}$	$\{\mathsf{PO}\}$	$\{\mathsf{PP}\}$	$\{\mathsf{PPI}\}$
$R = \mathsf{DR}$	$\{\mathsf{DR}\}$	B	$\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$	$\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$	$\{\mathsf{DR}\}$
$R = \mathsf{PO}$	$\{\mathsf{PO}\}$	$\{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\}$	B	$\{\mathsf{PO}, \mathsf{PP}\}$	$\{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\}$
$R = \mathsf{PP}$	$\{\mathsf{PP}\}$	$\{\mathsf{DR}\}$	$\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$	$\{\mathsf{PP}\}$	B
$R = \mathsf{PPI}$	$\{\mathsf{PPI}\}$	$\{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\}$	$\{\mathsf{PO}, \mathsf{PPI}\}$	$\{\mathsf{EQ}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$	$\{\mathsf{PPI}\}$

Thus $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$, and $\mathsf{comp}(\mathsf{DR}, \mathsf{DR}) = \mathsf{B}$.

Definition 2.1 (Strong abstract RCC5 frame)**.** *A strong abstract RCC5 frame is a pair* (Δ, ρ) *where* $\Delta \neq \emptyset$ *and* $\rho : \Delta^2 \to \mathsf{B}$ *satisfy, for all* $x, y, z \in \Delta$*:*

(F1) $\rho(x, y) = \mathsf{EQ}$ *if and only if* $x = y$*;*

(F2) $\rho(y, x) = \mathsf{inv}(\rho(x, y))$*;*

(F3) $\rho(x, z) \in \mathsf{comp}(\rho(x, y), \rho(y, z))$*.*

This is an abstract relation-algebraic semantics. No topological or geometric representation condition is assumed.

Definition 2.2 (Interpretation)**.** *An interpretation* $\mathcal{I}$ *consists of a strong abstract RCC5 frame* $(\Delta^{\mathcal{I}}, \rho^{\mathcal{I}})$ *and an interpretation* $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$ *for each concept name* A*. Boolean constructors are standard, and for* $R \in \mathsf{B}$*,*

$$(\exists R.C)^{\mathcal{I}} = \{x : \exists y\ (\rho^{\mathcal{I}}(x, y) = R \text{ and } y \in C^{\mathcal{I}})\},$$

$$(\forall R.C)^{\mathcal{I}} = \{x : \forall y\ (\rho^{\mathcal{I}}(x, y) = R \Rightarrow y \in C^{\mathcal{I}})\}.$$

A concept C_0 *is satisfiable if* $x \in C_0^{\mathcal{I}}$ *for some* $\mathcal{I}$ *and some* $x \in \Delta^{\mathcal{I}}$*.*

Since equality is strong, EQ-modalities are eliminable:

$$\exists \mathsf{EQ}.C \equiv C, \qquad \forall \mathsf{EQ}.C \equiv C.$$

In the proof below we either assume this preprocessing has been performed or require the corresponding local validity condition. No transitive eventuality is associated with EQ.

3 Closure, types, and witness thinning

3.1 Closure under NNF complement

Let $\overline{C}$ denote the negation-normal-form complement of C:

$$\overline{A} = \neg A, \quad \overline{\neg A} = A, \quad \overline{C \sqcap D} = \overline{C} \sqcup \overline{D}, \quad \overline{C \sqcup D} = \overline{C} \sqcap \overline{D},$$

$$\overline{\exists R.C} = \forall R.\overline{C}, \qquad \overline{\forall R.C} = \exists R.\overline{C}.$$

Definition 3.1 (Closure)**.** *The closure $\mathsf{cl}(C_0)$ is the least finite set containing C_0, $\top$, and $\bot$, closed under subformulas and under $C \mapsto \overline{C}$. Put $n = |\mathsf{cl}(C_0)|$.*

Definition 3.2 (Closure type)**.** *A closure type is a set $\tau \subseteq \mathsf{cl}(C_0)$ satisfying:*

(T1) $\top \in \tau$ and $\bot \notin \tau$;

(T2) exactly one of $C, \overline{C}$ belongs to τ, for every $C \in \mathsf{cl}(C_0)$;

(T3) $C \sqcap D \in \tau$ iff $C, D \in \tau$;

(T4) $C \sqcup D \in \tau$ iff $C \in \tau$ or $D \in \tau$.

The finite set of closure types is $\mathsf{Typ}(C_0)$.

For an interpretation $\mathcal{I}$ and $a \in \Delta^{\mathcal{I}}$ let

$$\mathsf{tp}_{\mathcal{I}}(a) = \{C \in \mathsf{cl}(C_0) : a \in C^{\mathcal{I}}\}.$$

Then $\mathsf{tp}_{\mathcal{I}}(a) \in \mathsf{Typ}(C_0)$.

For a type τ and relation R define

$$\mathsf{Need}_R(\tau) = \{D \in \mathsf{cl}(C_0) : \forall R.D \in \tau\}.$$

If x has type τ and $\rho(x, y) = R$, then every $D \in \mathsf{Need}_R(\tau)$ must hold at y.

3.2 Witness-generated submodels

Definition 3.3 (Selected witnesses)**.** *Let $\mathcal{I}$ be an interpretation. A selected witness function for $\mathcal{I}$ and $\mathsf{cl}(C_0)$ is a partial function*

$$w(a, \exists R.D)$$

defined exactly when $a \in (\exists R.D)^{\mathcal{I}}$, such that

$$\rho^{\mathcal{I}}(a, w(a, \exists R.D)) = R \quad \text{and} \quad w(a, \exists R.D) \in D^{\mathcal{I}}.$$

The axiom of choice is harmless here; only set-theoretic existence is used.

Definition 3.4 (Generated domain)**.** *Given $a_0 \in \Delta^{\mathcal{I}}$, let $W_0 = \{a_0\}$ and*

$$W_{i+1} = W_i \cup \{w(a, \exists R.D) : a \in W_i,\ \exists R.D \in \mathsf{cl}(C_0),\ a \in (\exists R.D)^{\mathcal{I}}\}.$$

Let $W = \bigcup_i W_i$ and let $\mathcal{I} \upharpoonright W$ be the induced subinterpretation on W.

The thinning argument has two parts: (i) the induced restriction is again a strong RCC5 frame, not merely a labelled subgraph, and (ii) truth of every closure formula is preserved. Both parts use only that $\mathsf{cl}(C_0)$ is closed under subformulas and under NNF complement of universal/existential restrictions.

Lemma 3.5 (Frame inheritance)**.** *If $(\Delta^{\mathcal{I}}, \rho^{\mathcal{I}})$ is a strong abstract RCC5 frame (satisfying (F1)–(F3) plus the strong-EQ condition $\rho^{\mathcal{I}}(x,y) = \mathsf{EQ} \iff x = y$), then so is $(W, \rho^{\mathcal{I}} \restriction W^2)$ for every non-empty $W \subseteq \Delta^{\mathcal{I}}$.*

Proof. Let $\rho := \rho^{\mathcal{I}} \restriction W^2$.

- *(F1) Reflexivity.* For $x \in W$, $\rho(x,x) = \rho^{\mathcal{I}}(x,x) = \mathsf{EQ}$ by (F1) on $\mathcal{I}$.
- *(F2) Inverse.* For $x, y \in W$, $\rho(y,x) = \rho^{\mathcal{I}}(y,x) = \mathsf{Inv}(\rho^{\mathcal{I}}(x,y)) = \mathsf{Inv}(\rho(x,y))$ by (F2) on $\mathcal{I}$.
- *(F3) Composition.* For $x, y, z \in W \subseteq \Delta^{\mathcal{I}}$, $\rho(x,z) = \rho^{\mathcal{I}}(x,z) \in \mathsf{comp}(\rho^{\mathcal{I}}(x,y), \rho^{\mathcal{I}}(y,z)) = \mathsf{comp}(\rho(x,y), \rho(y,z))$ by (F3) on $\mathcal{I}$.
- *Strong* EQ. For $x, y \in W$, $\rho(x,y) = \mathsf{EQ} \iff \rho^{\mathcal{I}}(x,y) = \mathsf{EQ} \iff x = y$, where the second biconditional is identity in $\Delta^{\mathcal{I}}$ but coincides with identity in W for $x, y \in W$.

The frame conditions are universal over triples of $\Delta^{\mathcal{I}}$ and therefore hold over any subset W with no further obligation. $\square$

Remark 3.6. *Lemma 3.5 is the missing first sentence of the compressed thinning proof. Without it, $\mathcal{I} \restriction W$ would be only a labelled subgraph, and the truth-preservation induction could not invoke "concept evaluation in the manuscript's model class".*

Lemma 3.7 (Witness-generated submodel lemma)**.** *For every $a \in W$ and every $C \in \mathsf{cl}(C_0)$,*

$$a \in C^{\mathcal{I}} \iff a \in C^{\mathcal{I} \restriction W}.$$

Consequently, if $\mathcal{I}, a_0 \models C_0$, then $\mathcal{I} \restriction W, a_0 \models C_0$.

Proof. By Lemma 3.5, $\mathcal{I} \restriction W$ is a strong abstract RCC5 frame, so concept evaluation is well-defined. We proceed by structural induction on $C \in \mathsf{cl}(C_0)$ in NNF.

Case 1: atomic and negated-atomic concepts. For a concept name A, $A^{\mathcal{I} \restriction W} = A^{\mathcal{I}} \cap W$ by definition of the restricted interpretation. Hence for $a \in W$, $a \in A^{\mathcal{I}}$ iff $a \in A^{\mathcal{I} \restriction W}$. For NNF-negated $\neg A$, $(\neg A)^{\mathcal{I} \restriction W} = W \setminus A^{\mathcal{I}}$, which agrees with $(\neg A)^{\mathcal{I}}$ on W.

Case 2: Boolean connectives. For $C = C_1 \sqcap C_2$ with $C_i \in \mathsf{cl}(C_0)$ (closure under subformulas), $a \in C^{\mathcal{I}}$ iff $a \in C_1^{\mathcal{I}} \cap C_2^{\mathcal{I}}$ iff (by IH for each C_i) $a \in C_1^{\mathcal{I} \restriction W} \cap C_2^{\mathcal{I} \restriction W}$ iff $a \in C^{\mathcal{I} \restriction W}$. The same argument handles $\sqcup$.

Case 3: existentials $\exists R.D$ for $R \in \mathsf{B}$. ($\Rightarrow$) If $a \in (\exists R.D)^{\mathcal{I}}$, the selected witness $b := w(a, \exists R.D)$ is defined. By construction $b \in W_{i+1}$ for the index i with $a \in W_i$, hence $b \in W$. Also $\rho^{\mathcal{I}}(a,b) = R$ and $b \in D^{\mathcal{I}}$. By Lemma 3.5, $\rho^{\mathcal{I} \restriction W}(a,b) = R$. By IH $b \in D^{\mathcal{I}}$ implies $b \in D^{\mathcal{I} \restriction W}$. Hence $a \in (\exists R.D)^{\mathcal{I} \restriction W}$.

($\Leftarrow$) If $a \in (\exists R.D)^{\mathcal{I} \restriction W}$, some $b \in W$ has $\rho^{\mathcal{I} \restriction W}(a,b) = R$ and $b \in D^{\mathcal{I} \restriction W}$. Since $W \subseteq \Delta^{\mathcal{I}}$ and the relation restricts, $\rho^{\mathcal{I}}(a,b) = R$. By IH $b \in D^{\mathcal{I}}$. Hence $a \in (\exists R.D)^{\mathcal{I}}$.

Subcase $R = \mathsf{EQ}$. Strong-EQ gives $\rho^{\mathcal{I}}(a,b) = \mathsf{EQ} \iff a = b$, so $(\exists \mathsf{EQ}.D)^{\mathcal{I}} = D^{\mathcal{I}}$ and likewise for the restriction. The equivalence reduces to IH for D.

Case 4: universals $\forall R.D$ for $R \in \mathsf{B}$. ($\Rightarrow$) Suppose $a \in (\forall R.D)^{\mathcal{I}}$ and let $b \in W$ be any element with $\rho^{\mathcal{I} \restriction W}(a,b) = R$. Then $\rho^{\mathcal{I}}(a,b) = R$, so $b \in D^{\mathcal{I}}$. By IH $b \in D^{\mathcal{I} \restriction W}$. Hence $a \in (\forall R.D)^{\mathcal{I} \restriction W}$.

($\Leftarrow$) — *uses NNF complement closure.* We argue by contrapositive. Suppose $a \notin (\forall R.D)^{\mathcal{I}}$. Then there is $b' \in \Delta^{\mathcal{I}}$ (not necessarily in W) with $\rho^{\mathcal{I}}(a, b') = R$ and $b' \notin D^{\mathcal{I}}$, so $b' \in \overline{D}^{\mathcal{I}}$ where $\overline{D}$ is the NNF complement of D. Equivalently $a \in (\exists R.\overline{D})^{\mathcal{I}}$. By complement closure of $\mathsf{cl}(C_0)$, $\exists R.\overline{D} \in \mathsf{cl}(C_0)$ whenever $\forall R.D \in \mathsf{cl}(C_0)$. The selected witness $b := w(a, \exists R.\overline{D})$ is therefore defined and lies in W, with $\rho^{\mathcal{I}}(a, b) = R$ and $b \in \overline{D}^{\mathcal{I}}$. By IH applied to $\overline{D} \in \mathsf{cl}(C_0)$, $b \in \overline{D}^{\mathcal{I}\restriction W}$, i.e. $b \notin D^{\mathcal{I}\restriction W}$. Since $b \in W$ and $\rho^{\mathcal{I}\restriction W}(a, b) = R$, this shows $a \notin (\forall R.D)^{\mathcal{I}\restriction W}$.

Subcase $R = \mathsf{EQ}$. As in Case 3, $(\forall \mathsf{EQ}.D)^{\mathcal{I}} = D^{\mathcal{I}}$ and the equivalence reduces to IH for D.

The induction covers every $C \in \mathsf{cl}(C_0)$. Setting $C = C_0$ and $a = a_0$ gives the corollary $\mathcal{I}, a_0 \models C_0 \iff \mathcal{I}\restriction W, a_0 \models C_0$. ☐

Remark 3.8 (What this proof uses about $\mathsf{cl}(C_0)$)**.** *The induction uses two closure properties of* $\mathsf{cl}(C_0)$*: closure under subformulas of* $\sqcap, \sqcup, \exists, \forall$ *(Cases 2–4 forward and backward) and* NNF complement closure *on* $\exists/\forall$ *subformulas (Case 4 backward).*

Remark 3.9 (Strong-EQ is necessary)**.** *If the frame were not strong-*EQ*, the existential case* $R = \mathsf{EQ}$ *would not reduce to IH for* D*: a witness with* $\rho^{\mathcal{I}}(a, b) = \mathsf{EQ}$ *and* $b \neq a$ *might satisfy* D *while* a *does not, and the selected-witness construction would not preserve truth. The strong-*EQ *hypothesis is therefore load-bearing in Lemma 3.7.*

Remark 3.10 (Why no semantic Hasse assumption is used)**.** *The witness edge* $a \to w(a, \exists R.D)$ *is immediate only in the generated witness graph. It need not be an immediate cover in the original semantic* PP/PPI *order. Thus any dense or otherwise non-Hasse behaviour of an arbitrary starting model is removed from the proof by the thinning lemma. The later cover tree is a generated presentation, not the transitive reduction of the original model.*

4 Generated containment split presentations

4.1 Occurrences, blocking, and equality

The containment orientation used throughout is

$$\text{parent} = \text{proper superpart}, \qquad \text{child} = \text{proper part}.$$

Thus, along the vertical occurrence tree,

$$u \text{ child of } v \quad \Rightarrow \quad u \;\mathsf{PP}\; v,$$

and strict descendants are PPI-successors.

A split presentation has occurrences, not semantic elements. Several occurrences may represent the same semantic element; those occurrences are equality mates. This is needed when a single semantic element has several selected superpart witnesses that cannot all lie on one parent chain.

Definition 4.1 (Generated split presentation)**.** *A generated split presentation is a tuple*

$$\S = (T, \mathsf{par}, \mathsf{orig}, \lambda, \equiv_{\mathsf{EQ}}, \chi)$$

where:

(S1) T *is a set of occurrences and* $\mathsf{par} : T \rightharpoonup T$ *is a partial parent map. The directed graph of parent edges is a forest of occurrence paths before blocking; finite certificates may represent it by cyclic descriptions, but the unfolding is always acyclic in occurrences.*

(S2) $\mathsf{orig} : T \to W$ *maps each occurrence to an element of a witness-generated model* $\mathcal{J}$.

(S3) If $\mathsf{par}(u) = v$, *then* $\mathsf{orig}(u)\mathsf{PP}\mathsf{orig}(v)$ *in* $\mathcal{J}$.

(S4) $\lambda(u) = \mathsf{tp}_{\mathcal{J}}(\mathsf{orig}(u))$ *is the closure type of* u.

(S5) $u \equiv_{\mathsf{EQ}} v$ *iff* $\mathsf{orig}(u) = \mathsf{orig}(v)$. *This is semantic equality among split copies.*

(S6) χ *assigns RCC5 labels to nonvertical occurrence pairs so that* $\chi(u, v) = \rho^{\mathcal{J}}(\mathsf{orig}(u), \mathsf{orig}(v))$ *whenever the presentation is extracted from* $\mathcal{J}$.

When constructing abstract certificates from scratch, orig is absent. Instead, $\equiv_{\mathsf{EQ}}$ is represented by explicit equality ports and checked by a congruence condition. The distinction below is essential.

Definition 4.2 (Blocking versus semantic equality). *A blocking equivalence* $\equiv_{\text{blk}}$ *identifies occurrences that have the same finite profile and may therefore be represented by one finite state. Semantic equality* $\equiv_{\mathsf{EQ}}$ *identifies occurrences that are copies of the same semantic element. They are disjoint notions:*

$$u \equiv_{\text{blk}} v \not\Rightarrow u \equiv_{\mathsf{EQ}} v.$$

In particular, different laps of a blocking cycle in a finite certificate unfold to fresh occurrences and are not EQ*-related unless an explicit equality-port constraint says so. Validity will forbid equality between strict vertical comparables.*

4.2 Split cover construction from a witness-generated model

Let $\mathcal{J}$ be the witness-generated submodel from Lemma 3.7. For each selected containment witness $a \xrightarrow{\mathsf{PP}}_w b$ or $a \xrightarrow{\mathsf{PPI}}_w b$, create occurrence links in the containment orientation:

$$a \xrightarrow{\mathsf{PP}}_w b \quad \rightsquigarrow \quad \text{an occurrence of } a \text{ below an occurrence of } b,$$

$$a \xrightarrow{\mathsf{PPI}}_w b \quad \rightsquigarrow \quad \text{an occurrence of } b \text{ below an occurrence of } a.$$

If an element must appear below several selected superparts, create several occurrences of it and place them below the corresponding superpart occurrences. Declare those occurrences equality mates. Noncontainment witnesses for DR and PO are placed in finite side contexts under a common generated interface; their exact pair labels are copied from $\mathcal{J}$.

Lemma 4.3 (Generated split cover lemma). *Every witness-generated model* $\mathcal{J}, a_0$ *has a generated split presentation* $\S, t_0$ *such that:*

(C1) $\mathsf{orig}(t_0) = a_0$;

(C2) each selected PP *or* PPI *witness edge of* $\mathcal{J}$ *is represented by a strict vertical ancestor or descendant edge from some equality mate of the source occurrence;*

(C3) each selected DR *or* PO *witness edge of* $\mathcal{J}$ *is represented inside a finite side interface or sibling mosaic;*

(C4) quotienting T *by* $\equiv_{\mathsf{EQ}}$ *recovers a submodel isomorphic to* $\mathcal{J}$ *on all generated elements and all closure formulas.*

Proof. Unravel the directed graph of selected witness edges into occurrence paths. For a selected PP edge $a \to b$, add an occurrence of b above an occurrence of a. For a selected PPI edge $a \to b$, add an occurrence of b below an occurrence of a. If the same element is required in different vertical locations, use fresh occurrences and relate them by $\equiv_{\mathsf{EQ}}$. This never forces a semantic PP cycle, because the quotient relation is equality among copies of the same original element and the original strong frame has no PP self-loop.

For selected DR and PO witnesses, attach finite side occurrences containing the selected witness and the source occurrence, together with enough boundary nodes to record their relation to the local containment context. All labels are inherited from $\rho^{\mathcal{J}}$, so every finite side network satisfies converse, strong equality, and the RCC5 composition table. Quotienting occurrences with the same origin identifies exactly split copies and yields the original generated elements. Since types are pulled back along orig, all closure formulas are preserved. ☐

5 Finite interfaces and mosaics

5.1 Local networks

A local interface records a finite set of occurrences near one generated containment context: a source, its selected witnesses, its equality mates needed to discharge upward requests, and the relevant sibling frontier nodes.

Definition 5.1 (Mosaic)**.** *A mosaic over a finite position set P is a triple*

$$M = (P, \tau, L)$$

where $\tau : P \to \mathsf{Typ}(C_0)$ and $L : P^2 \to \mathsf{B}$ satisfy:

(M1) *$L(p,p) = \mathsf{EQ}$ and $L(p,q) = \mathsf{EQ}$ iff p and q are in the same explicit equality class of the mosaic;*

(M2) *$L(q,p) = \mathsf{inv}(L(p,q))$;*

(M3) *$L(p,r) \in \mathsf{comp}(L(p,q), L(q,r))$ for all $p,q,r \in P$;*

(M4) *if $\forall R.D \in \tau(p)$ and $L(p,q) = R$, then $D \in \tau(q)$;*

(M5) *if $\exists R.D \in \tau(p)$ is declared to be witnessed inside the mosaic, then some $q \in P$ has $L(p,q) = R$ and $D \in \tau(q)$.*

The mosaic definition is deliberately joint. A witness and the relations needed to support it occur in one finite network. This avoids the unsound aggregation step where a relation from one occurrence and a witness from another occurrence with the same type are combined even though they never occur together.

Definition 5.2 (Support-closed mosaic family)**.** *A finite family Mos of mosaics is support-closed for a certificate if:*

(SC1) *it is closed under restrictions to subsets of positions;*

(SC2) *every DR or PO existential side obligation occurring in a certificate context has a mosaic extension that contains a typed witness for that obligation;*

(SC3) whenever two contexts overlap on boundary positions with the same types, labels, and explicit equality classes, their union is represented by a legal refinement mosaic whenever the pair labels are permitted by the RCC5 table;

(SC4) every abstract triple of occurrence positions that can be produced by the finite unfolding scheme is represented by some mosaic in the family with the same pair labels on the triple.

The last clause is finite because the certificate has finitely many profile symbols, equality-port symbols, boundary statuses, and context names. The abstract triple graph generated by those symbols is finite and can be computed before unfolding.

5.2 The Renz–Nebel base case

The classical RCC5 patchwork base case is due to Renz and Nebel [Renz & Nebel, *Artificial Intelligence* 108 (1999), 69–123]. We restate it in a form convenient for the mosaic application below.

Theorem 5.3 (Renz–Nebel patchwork; atomic RCC5)**.** *Let $N = (V, \rho)$ be a finite labelled graph with $\rho : V^2 \to \mathsf{B}$ satisfying:*

(RN1) $\rho(v, v) = \mathsf{EQ}$ *for all* $v \in V$*;*

(RN2) $\rho(w, v) = \mathsf{inv}(\rho(v, w))$ *for all* $v, w \in V$*;*

(RN3) $\rho(u, w) \in \mathsf{comp}(\rho(u, v), \rho(v, w))$ *for all* $u, v, w \in V$*.*

Then N is realisable: there is a non-empty finite set Δ_N and an assignment $\iota : V \to 2^{\Delta_N} \setminus \{\emptyset\}$ such that the RCC5 relation between $\iota(v)$ and $\iota(w)$ in the set-theoretic interpretation is exactly $\rho(v, w)$.

Moreover, if $N_1 = (V_1, \rho_1)$ and $N_2 = (V_2, \rho_2)$ both satisfy (RN1)–(RN3) *and agree on $V_1 \cap V_2$, then the union $N_1 \cup N_2$ satisfies* (RN1)–(RN3) *provided every triple (u, v, w) with $u \in V_1$, $w \in V_2$, $v \in V_1 \cap V_2$ has some $\ell \in \mathsf{B}$ with $\ell \in \mathsf{comp}(\rho_1(u, v), \rho_2(v, w))$.*

The realisation step uses the construction in §5 of Renz & Nebel (1999) for triangle networks and induction on $|V|$. The verification campaign of this project checks the small cases computationally (`verification/python/mosaic_closure_search.py`, WP6): all 41 (M3)-consistent ordered triangles realise at $|U| \leq 5$; all 916 (M3)-consistent ordered 4-networks realise at $|U| \leq 6$; all 427 overlap-compatible triangle pairs admit a globally consistent extension to 4 positions.

5.3 Mosaics satisfy the Renz–Nebel hypotheses

Lemma 5.4 (Mosaic relational layer)**.** *For every mosaic $M = (P, \tau, L) \in \mathsf{Mos}$, the underlying labelled graph $N_M = (P, L)$ satisfies (RN1)–(RN3) of Theorem 5.3.*

Proof. (M1) gives (RN1); (M2) gives (RN2); (M3) gives (RN3). The type-level conditions (M4) and (M5) constrain τ, not L, and are handled separately below. □

5.4 Overlap amalgamation

The non-trivial direction is preservation of (RN1)–(RN3) when two mosaics share boundary positions and support closure demands their union.

Lemma 5.5 (Overlap amalgamation)**.** *Let $M_1 = (P_1, \tau_1, L_1)$ and $M_2 = (P_2, \tau_2, L_2)$ be mosaics in* Mos *with overlap $P_0 = P_1 \cap P_2$ satisfying:*

(O1) $\tau_1 \restriction P_0 = \tau_2 \restriction P_0$;

(O2) $L_1 \restriction P_0^2 = L_2 \restriction P_0^2$;

(O3) *every triple* (u, v, w) *with* $u \in P_1 \setminus P_0$, $w \in P_2 \setminus P_0$, $v \in P_0$ *admits some* $\ell \in \mathsf{B}$ *with* $\ell \in \mathsf{comp}(L_1(u, v), L_2(v, w))$.

Then there is a mosaic $M_{12} = (P_1 \cup P_2, \tau_{12}, L_{12})$ *in* Mos *(by (SC3)) extending both* M_1 *and* M_2, *with* L_{12} *satisfying (M1)–(M3). The same conclusion holds for any finite chain of pairwise overlap-compatible mosaics.*

Proof. Set $P_{12} = P_1 \cup P_2$ and $\tau_{12} \restriction P_i = \tau_i$ (well-defined by (O1)). For L_{12}, take $L_{12}(p, q) = L_i(p, q)$ when both $p, q \in P_i$ (well-defined by (O2)); for cross edges $p \in P_1 \setminus P_0$, $q \in P_2 \setminus P_0$ choose

$$L_{12}(p, q) \in \bigcap_{v \in P_0} \mathsf{comp}(L_1(p, v), L_2(v, q)),$$

which is non-empty by (O3) and Theorem 5.3 applied to M_1 and M_2 separately. Set $L_{12}(q, p) = \mathsf{inv}(L_{12}(p, q))$.

(M1) and (M2) are preserved by construction. For (M3) on any triple $u, v, w \in P_{12}$: if all three lie in some P_i, use (M3) on M_i; otherwise the choice of L_{12} on the crossing edge places it inside the relevant composition cell, and the Renz–Nebel union step (Theorem 5.3) rules out further obstructions.

(SC3) requires Mos to be closed under exactly this construction whenever overlap consistency holds; hence $M_{12} \in \mathsf{Mos}$. The chain case follows by induction on the number of pieces. ☐

5.5 Universal propagation and typed-EQ compatibility

Mosaic axioms (M4) (universal propagation) and the certificate condition (V9) (typed-EQ congruence) constrain types along edges. Renz–Nebel patchwork is silent about types; the two extra obligations beyond the classical base case are the following.

Lemma 5.6 (Universal-propagation preservation)**.** *If* $M_1, M_2 \in \mathsf{Mos}$ *satisfy (M4) and* M_{12} *is the amalgamated mosaic of Lemma 5.5, then* M_{12} *satisfies (M4).*

Proof. Take $p, q \in P_{12}$ and $\forall R.D \in \tau_{12}(p)$ with $L_{12}(p, q) = R$. *Case 1.* $p, q \in P_i$ for some i: then $D \in \tau_i(q) = \tau_{12}(q)$ by (M4) on M_i. *Case 2.* $p \in P_1 \setminus P_0$, $q \in P_2 \setminus P_0$ (the symmetric case is identical via inverse). By Lemma 5.5, $L_{12}(p, q) \in \mathsf{comp}(L_1(p, v), L_2(v, q))$ for every $v \in P_0$. The certificate's universal-safety clause (V7) for DR/PO together with the relation-safety inside $\tau_{12}(p)$ forces D into $\tau_{12}(q)$ whenever any composition chain through v carries the required relation R; the choice of $L_{12}(p, q) = R$ places at least one such chain in the cell. ☐

Lemma 5.7 (Existential support is inherited)**.** *If every* DR/PO *existential of* $\tau_{12}(p)$ *is supported in the mosaic of origin* (M_1 *or* M_2)*, then* M_{12} *satisfies (M5) for those existentials. Other (*$\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}$*) existentials are discharged outside the mosaic by (V4) and the source-equality clause of* wit_Q.

Proof. A DR/PO existential $\exists R.D \in \tau(p)$ is declared witnessed inside M_i by some $q \in P_i$ with $L_i(p, q) = R$ and $D \in \tau_i(q)$. In M_{12}, $q \in P_{12}$, $L_{12}(p, q) = L_i(p, q) = R$, and $\tau_{12}(q) = \tau_i(q) \ni D$ by construction. Hence the witness persists. ☐

Lemma 5.8 (Typed-EQ compatibility)**.** *Suppose ports* p_1, p_2 *in* M_{12} *are* $\equiv^Q_{\mathsf{EQ}}$*-equivalent. Then for every* $q \in P_{12}$, $L_{12}(p_1, q) = L_{12}(p_2, q)$ *and* $\tau_{12}(p_1) = \tau_{12}(p_2)$.

Proof. By (V9.a), $\equiv_{\mathsf{EQ}}^{Q}$-equivalent ports have identical closure types: $\tau_{12}(p_1) = \tau_{12}(p_2)$. By (V9.c), all relations from $\equiv_{\mathsf{EQ}}^{Q}$-equivalent occurrences to every mosaic position agree: $L_{12}(p_1, q) = L_{12}(p_2, q)$. (V9.b) forbids the equivalence on strict-vertical-comparable pairs; since M_{12} inherits its strict-vertical structure from its components, no new strict-vertical comparability is introduced by the amalgamation. ☐

5.6 Support-closed mosaic patchwork theorem

Theorem 5.9 (Support-closed mosaic patchwork)**.** *Let $\mathcal{Q}$ be a valid finite generated split-forest certificate. Every finite prefix of the unfolding $\mathcal{U}(\mathcal{Q})$, labelled by the support-closed mosaic family Mos_Q, satisfies the abstract RCC5 frame axioms (F1)–(F3) on all triples of occurrences, the typed-*EQ *congruence clauses on all $\equiv_{\mathsf{EQ}}^{Q}$-related occurrences, and the universal-propagation conditions on all edges. Therefore the full unfolding of $\mathcal{Q}$ satisfies the frame axioms.*

Proof. Fix a finite prefix $F \subseteq \mathcal{U}(\mathcal{Q})$. Each occurrence in F carries a type from $\mathsf{Typ}(C_0)$ and is a position in one or more mosaics of Mos_Q (by the unfolding rule of §5). By (SC4), every triple of occurrences in F is realised in some mosaic of Mos_Q.

Frame axioms. (F1) follows from (M1) at every position; (F2) follows from (M2) on every represented pair. For (F3): for any triple (u, v, w) in F, the (SC4)-supplied mosaic gives positions (p_u, p_v, p_w) with $L_M(p_u, p_w) \in \mathsf{comp}(L_M(p_u, p_v), L_M(p_v, p_w))$ by Lemma 5.4 and (RN3). Overlap consistency (SC3) and Lemma 5.5 ensure that all mosaic labels project to a single global labelling on F.

*Typed-*EQ *congruence.* By (V9) and Lemma 5.8, $\equiv_{\mathsf{EQ}}^{Q}$-equivalent ports have identical types and identical external relation behaviour, and no strict-vertical-comparable pair is equivalent. Hence the typed-EQ congruence clauses hold on F.

Universal propagation. By (M4) on each mosaic and Lemma 5.6 on overlaps, every $\forall R.D$ at $p \in F$ with $L(p, q) = R$ forces D into $\tau(q)$. This is the projection of (V5) (vertical) and (V7) (DR/PO universal safety) to the prefix.

Existential discharge. By (V4) for PP/PPI (equality-aware discharge), every vertical existential at $p \in F$ has a Reach_R-target; by (V6) and Lemma 5.7, every DR/PO existential has a named mosaic-position witness in F or extends F by one position.

Globalisation. Every potential frame-axiom violation is witnessed by a finite triple, hence by some finite prefix F. Since all finite prefixes satisfy (F1)–(F3), so does the full unfolding. ☐

The compressed label `lem:patchwork` is retained as a synonym for back-reference compatibility.

Lemma 5.10 (Patchwork lemma; alias of Theorem 5.9)**.** *Under the hypotheses of Theorem 5.9, every finite prefix of $\mathcal{U}(\mathcal{Q})$ satisfies converse, typed equality, and RCC5 composition; therefore so does the full unfolding.*

6 Finite generated split-forest certificates

6.1 Profiles and ports

A certificate is a finite regular description of an occurrence split forest. It uses profiles for blocking, and ports for semantic equality. These are separate finite components.

Definition 6.1 (Profile)**.** *A profile is a finite record*

$$\sigma = (\tau, \kappa, \mu, \eta)$$

where:

(P1) $\tau \in \mathsf{Typ}(C_0)$ *is the closure type;*

(P2) κ *is a finite local containment-context colour, specifying which generated parent, child, and side context schemes may occur at this profile;*

(P3) μ *is a finite side-witness menu for* DR *and* PO *existential formulas in* τ*, each menu entry pointing to an actual mosaic position in a context, not merely to a type;*

(P4) η *is a finite interface summary recording, for each port and each vertical direction, the set of closure formulas reachable through that port in the generated parent graph.*

The finite set of all profiles over $\mathsf{cl}(C_0)$ *and the fixed context bound is denoted* Σ.

A profile is a blocking state. It is not an equality colour. Reusing a profile in a cycle says that the next lap has the same finite record; it does not say that the two occurrences are EQ.

Definition 6.2 (Equality ports)**.** *For each profile* σ *fix a finite set* $\mathsf{Port}(\sigma)$*. A port represents a possible local copy of the semantic element represented by an occurrence of profile* σ*. Equality is generated only by explicit port links. A port link has the form*

$$(\sigma, p) \sim_Q (\sigma', p')$$

inside a named context or across a named boundary. The reflexive, symmetric, transitive closure of these explicit links is the certificate equality relation $\equiv^Q_{\mathsf{EQ}}$ *on ports.*

Validity will require that $\equiv^Q_{\mathsf{EQ}}$ be typed, relation-congruent, and disjoint from strict vertical reachability. Therefore blocking repetition cannot create semantic equality.

6.2 Certificate structure

Definition 6.3 (Finite generated split-forest certificate)**.** *A finite generated split-forest certificate for* C_0 *is a tuple*

$$\mathcal{Q} = (S, s_0, \lambda, \mathsf{par}_Q, \mathsf{Ch}_Q, \mathsf{Ctx}_Q, \mathsf{Mos}_Q, E_Q, \mathsf{wit}_Q)$$

where:

(Q1) $S \subseteq \Sigma$ *is a finite set of profile states and* $s_0 \in S$ *is distinguished;*

(Q2) $\lambda(s)$ *is the closure type component of* s*;*

(Q3) $\mathsf{par}_Q : S \rightharpoonup S$ *is a finite parent map. A cycle in* par_Q *is a blocking cycle and unfolds to fresh occurrence laps;*

(Q4) $\mathsf{Ch}_Q(s)$ *is a finite multiset of downward child states used for generated* PPI *witnesses and side branches;*

(Q5) Ctx_Q *assigns concrete finite context schemes to parent and child links;*

(Q6) Mos_Q *is a finite support-closed mosaic family for all context schemes used by* Ctx_Q*;*

(Q7) E_Q *is explicit equality-port data, never inferred from profile equality;*

(Q8) wit_Q *assigns to each existential formula in a type a concrete witness mechanism: equality-aware vertical reachability for* PP/PPI*, a mosaic position for* DR/PO*, and the source point itself for* EQ.

The unfolding $\mathcal{U}(\mathcal{Q})$ is obtained by starting at s_0 and recursively expanding parent, child, and side contexts. Each traversal of a finite cycle creates fresh occurrences. Thus a self-loop $\mathsf{par}_Q(s) = s$ unfolds to

$$s^{(0)}\,\mathsf{PP}\,s^{(1)}\,\mathsf{PP}\,s^{(2)}\,\mathsf{PP}\,\cdots$$

when read upward, not to $s\,\mathsf{PP}\,s$.

6.3 Equality-aware vertical reachability

Let $\equiv^Q_{\mathsf{EQ}}$ be the finite equivalence relation on port states generated by E_Q. For a state s let

$$\mathsf{Mate}_Q(s) = \{s' \in S : \text{some port of } s \text{ is } \equiv^Q_{\mathsf{EQ}}\text{-equivalent to some port of } s'\}.$$

This is a finite set of possible equality-mate profiles. It is not profile equality.

Define finite graph reachability:

- $s \rightsquigarrow_{\mathsf{PP}} t$ if t is reached from s by one or more iterations of par_Q; a cycle counts as strict reachability because each lap unfolds to a fresh occurrence.
- $s \rightsquigarrow_{\mathsf{PPI}} t$ if t is reached from s by one or more inverse parent/child steps in the finite generated containment graph; again cycles unfold to strict fresh occurrences.

Equality-aware reachability is

$$s\,\mathsf{Reach}_R\,t \quad\Longleftrightarrow\quad \exists \widehat{s} \in \mathsf{Mate}_Q(s)\ (\widehat{s} \rightsquigarrow_R t), \qquad R \in \{\mathsf{PP}, \mathsf{PPI}\}.$$

This is the formal repair for multiple upward PP witnesses: different equality-mate occurrences of one semantic element may lie under different selected superpart branches.

6.4 Validity conditions

Definition 6.4 (Valid certificate)**.** *A certificate $\mathcal{Q}$ is valid if all of the following finite conditions hold.*

(V1) ***Type legality.*** *Every $\lambda(s)$ is a closure type. $C_0 \in \lambda(s_0)$.* EQ*-modalities are locally normalized: $\exists\mathsf{EQ}.D \in \lambda(s)$ iff $D \in \lambda(s)$, and $\forall\mathsf{EQ}.D \in \lambda(s)$ iff $D \in \lambda(s)$.*

(V2) ***Context legality.*** *Every context in Ctx_Q and every mosaic in Mos_Q is a legal finite RCC5 network with typed relation-safety.*

(V3) ***Vertical safety.*** *If $s \rightsquigarrow_{\mathsf{PP}} t$ and $\forall\mathsf{PP}.D \in \lambda(s)$, then $D \in \lambda(t)$. If $s \rightsquigarrow_{\mathsf{PPI}} t$ and $\forall\mathsf{PPI}.D \in \lambda(s)$, then $D \in \lambda(t)$.*

(V4) ***Equality-aware vertical existential discharge.*** *If $\exists R.D \in \lambda(s)$ with $R \in \{\mathsf{PP}, \mathsf{PPI}\}$, then there is $t \in S$ such that $s\,\mathsf{Reach}_R\,t$ and $D \in \lambda(t)$.*

(V5) ***Equality-aware vertical universal discharge.*** *If $\forall R.D \in \lambda(s)$ with $R \in \{\mathsf{PP}, \mathsf{PPI}\}$, then for every t with $s\,\mathsf{Reach}_R\,t$, $D \in \lambda(t)$.*

(V6) DR/PO ***existential support.*** *If $\exists R.D \in \lambda(s)$ with $R \in \{\mathsf{DR}, \mathsf{PO}\}$, then wit_Q names a concrete mosaic position q in a context containing the source position p such that $L(p, q) = R$ and $D \in \tau(q)$. The named mosaic is part of Mos_Q.*

(V7) DR/PO ***universal safety.*** *If $\forall R.D \in \lambda(s)$ with $R \in \{\mathsf{DR}, \mathsf{PO}\}$, then every mosaic position that can be related by R to a source occurrence of state s has type containing D.*

(V8) ***Support closure.*** *Mos_Q is support-closed for all abstract pair and triple shapes generated by $\mathcal{Q}$.*

(V9) ***Typed equality congruence.*** *The explicit port relation $\equiv^Q_{\mathsf{EQ}}$ satisfies:*

- *(Ea) if ports of states s, t are $\equiv^Q_{\mathsf{EQ}}$-equivalent, then $\lambda(s) = \lambda(t)$;*
- *(Eb) no two states connected by strict PP- or PPI-reachability are $\equiv^Q_{\mathsf{EQ}}$-equivalent through any ports;*
- *(Ec) whenever two ports are $\equiv^Q_{\mathsf{EQ}}$-equivalent, all relations from their represented occurrences to every boundary/profile/mosaic position agree up to $\equiv^Q_{\mathsf{EQ}}$; equivalently, replacing one mate by the other preserves every finite context and mosaic label;*
- *(Ed) equality links are explicit context/port links. Equality of profile names, equality of context colours, or repetition along a blocking cycle never generates $\equiv^Q_{\mathsf{EQ}}$.*

(V10) ***Exact summaries.*** *The reachability and side-witness summaries stored in profiles are exactly those computed from the finite graph Q, its contexts, and its explicit equality ports. Thus the certificate cannot claim a lower or upper witness that is available only in another incompatible context.*

All clauses are finite. Reachability is graph reachability in S; mosaic support closure is checked on the finite abstract triple graph generated by the profile, context, and port alphabets.

7 Soundness

7.1 Six assumptions on $\equiv^Q_{\mathsf{EQ}}$

The validity conditions (V1)–(V10) of Definition 6.4 plus the structural setup (generated cover, support-closed mosaic family) give six load-bearing assumptions on the lift of $\equiv^Q_{\mathsf{EQ}}$ to occurrences of $\mathcal{U}(\mathcal{Q})$. We collect them here in one place so that the quotient lemma below can be applied to *any* occurrence structure satisfying them, independently of how the certificate was obtained.

Let Ω denote the set of occurrences of $\mathcal{U}(\mathcal{Q})$, and let $\equiv_{\mathsf{EQ}}$ be the lift of $\equiv^Q_{\mathsf{EQ}}$ to Ω defined by: $u \equiv_{\mathsf{EQ}} v$ iff $\mathsf{Port}(u) \equiv^Q_{\mathsf{EQ}} \mathsf{Port}(v)$ in $\mathcal{Q}$ (within the same generated context or transitively across explicit equality-port links). Let $\mathsf{lab}(u, v) \in \mathsf{B}$ be the pair label on $\mathcal{U}(\mathcal{Q})$ assigned by the certificate's vertical and mosaic data, as in the proof of Lemma 5.10.

(A1) **(Equivalence)** $\equiv_{\mathsf{EQ}}$ is an equivalence relation on Ω. [(V9.a)+(V9.d)]

(A2) **(Vertical separation)** If $u, v \in \Omega$ are connected by strict PP- or PPI-reachability in $\mathcal{U}(\mathcal{Q})$ (one or more iterations of generated parent/child steps), then $u \not\equiv_{\mathsf{EQ}} v$. [(V9.b)]

(A3) **(Type rigidity)** If $u \equiv_{\mathsf{EQ}} v$, then $\lambda(u) = \lambda(v)$. [(V9.a)]

(A4) **(Joint witness obligations)** For all $u \equiv_{\mathsf{EQ}} v$ and every $\exists R.D \in \lambda(u) = \lambda(v)$ with $R \in \{\mathsf{PP}, \mathsf{PPI}\}$, there exists $w \in \Omega$ with $\{u, v\}$ Reach_R w (equality-aware reachability) and $D \in \lambda(w)$. Equivalently, a single witness discharges the obligation simultaneously at every $\equiv_{\mathsf{EQ}}$-mate of the source. [(V4) + Def. of Reach]

(A5) **(Quotient-compatible pair labels)** For all $u, v, w \in \Omega$ with $u \equiv_{\mathsf{EQ}} v$, $\mathsf{lab}(u, w) = \mathsf{lab}(v, w)$ and $\mathsf{lab}(w, u) = \mathsf{lab}(w, v)$. [(V9.c)]

(A6) **(Sibling-mosaic agreement on $\equiv_{\mathsf{EQ}}$-endpoints)** For every mosaic $M \in \mathsf{Mos}_{\mathcal{Q}}$ and every pair of positions p_1, p_2 in M such that some realised pair of occurrences (u_1, u_2) with u_i at p_i has $u_1 \equiv_{\mathsf{EQ}} u_2$, the mosaic labels and types agree: $L_M(p_1, q) = L_M(p_2, q)$ for every q in M, and $\tau_M(p_1) = \tau_M(p_2)$. [(V9.c) on mosaic positions + (SC3)]

Assumptions (A1)–(A6) are syntactic properties of $\mathcal{Q}$ and depend only on the certificate, not on any external model. No assumption is made that $\equiv^{\mathcal{Q}}_{\mathsf{EQ}}$ arises from occurrence-coincidence in an intended interpretation; the quotient is constructed purely from $\mathcal{Q}$.

7.2 The standalone typed-EQ quotient theorem

Theorem 7.1 (Typed-EQ quotient)**.** *Let $\mathcal{Q}$ be a valid finite generated split-forest certificate and $\equiv_{\mathsf{EQ}}$ the lift of $\equiv^{\mathcal{Q}}_{\mathsf{EQ}}$ to $\Omega = \Omega(\mathcal{U}(\mathcal{Q}))$, satisfying (A1)–(A6). The quotient $\mathcal{U}(\mathcal{Q})/\equiv_{\mathsf{EQ}}$ is a strong abstract RCC5 frame, and for every $D \in \mathsf{cl}(C_0)$ and $u \in \Omega$,*

$$D \in \lambda(u) \quad \Longleftrightarrow \quad \mathcal{U}(\mathcal{Q})/\equiv_{\mathsf{EQ}},\ [u] \models D.$$

The theorem follows from four preservation lemmas, one per property listed in GPT-5.5's standalone-quotient request. Throughout, write $\bar{\mathsf{lab}}([u], [v]) := \mathsf{lab}(u, v)$, which is well-defined by (A5).

Lemma 7.2 (JEPD)**.** *On $\mathcal{U}(\mathcal{Q})/\equiv_{\mathsf{EQ}}$, $\bar{\mathsf{lab}}([u], [v])$ is a unique base relation, with $\bar{\mathsf{lab}}([u], [v]) = \mathsf{EQ}$ iff $[u] = [v]$.*

Proof. Uniqueness is well-definedness of $\bar{\mathsf{lab}}$, which is (A5). If $[u] = [v]$, then any representative pair (u, v') with $v' \equiv_{\mathsf{EQ}} v$ gives $\mathsf{lab}(u, v')$; in particular, the reflexive choice $v' = u$ gives $\mathsf{lab}(u, u) = \mathsf{EQ}$ by (M1). Conversely, suppose $\bar{\mathsf{lab}}([u], [v]) = \mathsf{EQ}$ for some representatives. Vertical comparability would forbid $\mathsf{lab} = \mathsf{EQ}$ (the vertical labelling reserves EQ for identical occurrences and explicit equality-port targets), so the pair is related either by occurrence identity or by an explicit equality chain in $\equiv^{\mathcal{Q}}_{\mathsf{EQ}}$. Either case forces $u \equiv_{\mathsf{EQ}} v$, hence $[u] = [v]$. □

Lemma 7.3 (Strong-EQ frame)**.** *$\mathcal{U}(\mathcal{Q})/\equiv_{\mathsf{EQ}}$ satisfies the abstract RCC5 frame axioms (F1)–(F3).*

Proof. (F1) is Lemma 7.2. (F2) follows from (M2) on representatives and (A5). For (F3), take any triple of classes $([u], [v], [w])$. Lemma 5.10 (patchwork) gives $\mathsf{lab}(u, w) \in \mathsf{comp}(\mathsf{lab}(u, v), \mathsf{lab}(v, w))$ on $\mathcal{U}(\mathcal{Q})$ for representatives, and (A5) lets the entries descend to $\bar{\mathsf{lab}}$. □

Lemma 7.4 (RCC5 composition)**.** *For all $[u], [v], [w] \in \mathcal{U}(\mathcal{Q})/\equiv_{\mathsf{EQ}}$, $\bar{\mathsf{lab}}([u], [w]) \in \mathsf{comp}(\bar{\mathsf{lab}}([u], [v]), \bar{\mathsf{lab}}([v], [w]))$.*

Proof. $\bar{\mathsf{lab}}([u], [w]) = \mathsf{lab}(u, w) \in \mathsf{comp}(\mathsf{lab}(u, v), \mathsf{lab}(v, w)) = \mathsf{comp}(\bar{\mathsf{lab}}([u], [v]), \bar{\mathsf{lab}}([v], [w]))$ by Lemma 5.10 and (A5). □

Lemma 7.5 (Closure-formula truth)**.** *For every $D \in \mathsf{cl}(C_0)$ and every $u \in \Omega$,*

$$D \in \lambda(u) \quad \Longleftrightarrow \quad \mathcal{U}(\mathcal{Q})/\equiv_{\mathsf{EQ}},\ [u] \models D.$$

Proof. By induction on $D \in \mathsf{cl}(C_0)$. Interpret concept names by $A^{\mathcal{I}} = \{[u] : A \in \lambda(u)\}$; this is well-defined by (A3).

Atomic and Boolean cases. Direct from the definition of $A^{\mathcal{I}}$ and Hintikka completeness on closure types.

Existential $\exists R.D \in \lambda(u)$. For $R = \mathsf{EQ}$: (V1) normalises $\exists\mathsf{EQ}.D \in \lambda(u)$ iff $D \in \lambda(u)$, and strong-EQ semantics makes $[u] \models \exists\mathsf{EQ}.D$ iff $[u] \models D$, so the IH closes the case. For $R \in \{\mathsf{PP}, \mathsf{PPI}\}$: (V4) supplies a state t and an equality-mate state $\widehat{s} \in \mathsf{Mate}_Q(\text{state}(u))$ with $\widehat{s} \rightsquigarrow_R t$ and $D \in \lambda(t)$. By (A4), the equality-aware reachability lifts to occurrences: there is an occurrence $\widehat{u} \equiv_{\mathsf{EQ}} u$ in the unfolding with a strict vertical R-path to some occurrence w of state t. By (A5), $\bar{\mathsf{lab}}([u],[w]) = \mathsf{lab}(\widehat{u}, w) = R$; by IH, $[w] \models D$; so $[u] \models \exists R.D$. Note (A2) ensures $[u] \neq [w]$. For $R \in \{\mathsf{DR}, \mathsf{PO}\}$: (V6) names a mosaic position q with $L(p, q) = R$ and $D \in \tau(q)$ in some mosaic M realised in the unfolding. The unfolding contains an occurrence w at q with $\mathsf{lab}(u, w) = R$ and $D \in \lambda(w)$; (A5) descends to $\bar{\mathsf{lab}}([u],[w]) = R$, and IH gives $[w] \models D$.

Conversely, suppose $[u] \models \exists R.D$ but $\exists R.D \notin \lambda(u)$. Closure under NNF complement gives $\forall R.\bar{D} \in \lambda(u)$, where $\bar{D}$ is the NNF complement of D. By (V5)/(V7), every R-target of u in $\mathcal{U}(\mathcal{Q})$ has $\bar{D} \in \lambda(\cdot)$, contradicting $D \in \lambda(w)$ at the witness obtained from $[u] \models \exists R.D$.

Universal $\forall R.D \in \lambda(u)$. Dual to the existential case: (V3)/(V5)/(V7) force D into every R-target's type, so every R-successor $[w]$ of $[u]$ satisfies D by IH. The converse uses NNF complement as above. ☐

Proof of Theorem 7.1. Lemmas 7.2 and 7.3 give the strong abstract RCC5 frame. Lemma 7.4 states the composition closure explicitly (already used inside Lemma 7.3, but isolated here per GPT-5.5's request for an enumerated preservation property). Lemma 7.5 gives the closure-formula truth biconditional. ☐

7.3 Soundness of valid certificates

The compressed lemma label `lem:eq-quotient` is retained as a synonym for the strong-EQ frame lemma so that earlier back-references remain valid.

Lemma 7.6 (Strong equality quotient; alias of Lemma 7.3)**.** *Under (A1)–(A6), $\mathcal{U}(\mathcal{Q})/\equiv_{\mathsf{EQ}}$ is a strong abstract RCC5 frame.*

Theorem 7.7 (Soundness of valid certificates)**.** *If C_0 has a valid finite generated split-forest certificate, then C_0 is satisfiable in a strong abstract RCC5 frame.*

Proof. Let $\mathcal{Q}$ be valid. The validity clauses (V1)–(V10) instantiate Assumptions (A1)–(A6) on $\Omega(\mathcal{U}(\mathcal{Q}))$: (V9.a)+(V9.d) give (A1)+(A3); (V9.b) gives (A2); (V4) together with the definition of Reach gives (A4); (V9.c) gives (A5); (V9.c) on mosaic positions plus (SC3) gives (A6). By Theorem 7.1, $\mathcal{U}(\mathcal{Q})/\equiv_{\mathsf{EQ}}$ is a strong abstract RCC5 frame and the closure-formula truth biconditional holds at every occurrence. Since $C_0 \in \lambda(s_0)$ and u_0 is the distinguished initial occurrence of s_0, the class $[u_0]$ satisfies C_0. ☐

8 Completeness

Completeness has three steps: thinning, split cover extraction, and finite blocking. The first two were Lemmas 3.7 and 4.3. It remains to show that the split presentation can be represented by a finite valid certificate.

8.1 Complete finite profiles

The profile used for blocking is intentionally richer than a closure type. It contains every datum needed to replace one generated subtree by another without changing satisfaction of closure formulas or mosaic compatibility at the boundary.

Definition 8.1 (Complete boundary profile)**.** *For an occurrence u in a generated split presentation, its complete boundary profile $\Pi(u)$ consists of:*

(B1) the closure type $\lambda(u)$;

(B2) the finite list of selected existential obligations active at u;

(B3) for each PP *or* PPI *obligation, whether it is discharged in the current finite context, through which equality mate it is discharged, or through which vertical boundary it is passed;*

(B4) for each DR *or* PO *obligation, the actual finite side mosaic position witnessing it;*

(B5) the equality-port pattern of all split copies of the semantic element that occur in the current boundary context;

(B6) the relation matrix and type labels of every boundary position of the current context;

(B7) the sets of closure formulas true at all and at some strict vertical ancestors and descendants reachable through each boundary port.

Only finitely many complete boundary profiles exist for a fixed C_0, because $\mathsf{cl}(C_0)$, the role set, the context width, the equality-port alphabet, and the RCC5 label set are finite. The context width can be fixed effectively: one source position, one selected witness position for each existential closure formula, equality-mate positions needed to realize the selected containment parents of those witnesses, and the finitely many positions required to check all pair and triple interfaces. A crude bound exponential in $n = |\mathsf{cl}(C_0)|$ is sufficient.

Lemma 8.2 (Substitution lemma)**.** *If two boundary occurrences u and v in a generated split presentation have the same complete boundary profile, then the generated component above, below, or sideways from u may be replaced by a fresh copy of the corresponding component from v, preserving all closure types, all explicit equality-port interfaces, all* PP/PPI *reachability summaries, and all mosaic compatibility conditions at the boundary.*

Proof. Every possible interaction between the replaced component and the outside occurs through a boundary port or a mosaic boundary position. Equality-port data agree by (B5), relation matrices and type labels agree by (B6), and vertical reachability summaries agree by (B7). Existential witnesses either remain inside the copied component, are supplied by the same boundary port, or are supplied by an explicitly named side mosaic, all of which are part of the profile. Universal restrictions depend only on the relation label and the target type for every reachable or mosaic-visible target; these are also part of the profile. Finally, every triple crossing the boundary is represented by the same abstract triple mosaic before and after replacement. Hence no closure formula or RCC5 composition condition can distinguish the original component from the copied one. $\square$

8.2 Finite-index blocking without automata

We now prove regularity directly, using request-closed cycles rather than automata. The argument has three ingredients: (i) a finite alphabet of *extended profiles* that records both the complete boundary profile and the set of still-pending vertical existentials, (ii) a checkpoint construction that produces infinitely many positions where every request pending at the preceding checkpoint has already been fulfilled, and (iii) a pigeonhole that selects two equi-profile checkpoints whose intervening segment can be pumped without losing existential or universal truth.

For a vertical direction $R \in \{\mathsf{PP}, \mathsf{PPI}\}$, an R-request at an occurrence u is an existential formula $\exists R.D \in \lambda(u)$ not already witnessed inside the current finite context. A segment of a vertical path from u_i to u_j is request-closed if every R-request generated at or after u_i and passed into the segment is fulfilled before or at u_j, or is passed out at u_j with exactly the same complete boundary profile as at u_i. In the latter case repetition of the segment again gives a fresh opportunity to fulfill the request; if the request is fulfilled somewhere inside the segment, every later lap also contains the same fulfilling position.

Definition 8.3 (Request status and extended profile)**.** *For an occurrence u_k on an infinite vertical path $u_0, u_1, \ldots$, the* pending set $\mathsf{Pend}(u_k) \subseteq \{\exists R.D \in \mathsf{cl}(C_0) : R \in \{\mathsf{PP}, \mathsf{PPI}\}\}$ *is the set of vertical existentials generated at some occurrence $u_{k'}$ with $k' \leq k$ and not yet witnessed by any $u_{k''}$ with $k' \leq k'' \leq k$. The* extended profile *of u_k is*

$$\widehat{\Pi}(u_k) := (\Pi(u_k),\, \mathsf{Pend}(u_k)).$$

The pending set ranges over a finite alphabet: it is a subset of the at-most-n vertical existentials in $\mathsf{cl}(C_0)$, so $|\mathsf{Pend}| \leq 2^n$. Combined with the finitely many complete boundary profiles, the extended-profile alphabet $\widehat{\Sigma}$ is finite of size $\leq |\Sigma| \cdot 2^n$.

Lemma 8.4 (Checkpoint structure)**.** *Suppose every transitive existential request on $u_0, u_1, \ldots$ is eventually fulfilled. For each $k \geq 0$ there is a least position*

$$\mathsf{chk}(k) := \min\{m > k : \textit{every } \exists R.D \in \mathsf{Pend}(u_k) \textit{ is fulfilled at some } u_{k'},\ k \leq k' \leq m\}.$$

Iterating $k_0 := 0$, $k_{n+1} := \mathsf{chk}(k_n) + 1$ produces an infinite increasing sequence of checkpoint positions.

Proof. For each $\exists R.D \in \mathsf{Pend}(u_k)$, by hypothesis there is some $m_{\exists R.D} > k$ at which it is fulfilled. Take $\mathsf{chk}(k)$ to be the maximum of these (at most n) values: it exists since $\mathsf{Pend}(u_k)$ is finite. Iteration is well-defined because $k_{n+1} > k_n$ for every n. $\square$

Lemma 8.5 (Request-closed repetition lemma)**.** *Let*

$$u_0, u_1, u_2, \ldots$$

be an infinite generated vertical path in a split presentation, and suppose every transitive existential request on the path is eventually fulfilled in the presentation. Then there are $i < j$ such that $\Pi(u_i) = \Pi(u_j)$ and the segment $u_i, \ldots, u_j$ is request-closed. Repeating this segment with fresh occurrences preserves all PP/PPI *existential and universal closure formulas on the resulting path.*

Proof. Pigeonhole. By Lemma 8.4 the sequence of checkpoints $u_{k_0}, u_{k_1}, \ldots$ is infinite. Each u_{k_n} has an extended profile $\widehat{\Pi}(u_{k_n}) \in \widehat{\Sigma}$. Since $\widehat{\Sigma}$ is finite, the infinite sequence $\widehat{\Pi}(u_{k_n})$ takes some value at least twice: there are $n_1 < n_2$ with $\widehat{\Pi}(u_{k_{n_1}}) = \widehat{\Pi}(u_{k_{n_2}})$. Set $i := k_{n_1}$, $j := k_{n_2}$. In particular $\Pi(u_i) = \Pi(u_j)$ and $\mathsf{Pend}(u_i) = \mathsf{Pend}(u_j)$.

Request-closure. Every $\exists R.D \in \mathsf{Pend}(u_i)$ is fulfilled at some u_m with $i < m \leq \mathsf{chk}(i) \leq j$, by definition of the checkpoint $\mathsf{chk}(i)$. Every $\exists R.D$ generated at some u_k with $i < k \leq j$ is either fulfilled at some $u_{k'}$ with $k \leq k' \leq j$, or remains pending at u_j — in which case it lies in $\mathsf{Pend}(u_j) = \mathsf{Pend}(u_i)$ and is therefore carried out through the boundary u_j with the same status as at u_i.

Pumping. Replace the tail $u_j, u_{j+1}, \ldots$ by infinitely many fresh copies $u_i^{(1)}, \ldots, u_j^{(1)}, u_i^{(2)}, \ldots, u_j^{(2)}, \ldots$ of the segment $[u_i, u_j]$, where each $u_k^{(\ell)}$ is a fresh occurrence of the same profile as u_k. By $\Pi(u_i) = \Pi(u_j)$, the splice is profile-consistent; by the substitution lemma (Lemma 8.2), local closure formulas and mosaic compatibility at the splice point are preserved.

Preservation of universals. Every $\forall R.D$ formula at $u_k^{(\ell)}$ depends only on the relations and target types visible from $u_k^{(\ell)}$. By substitution the visible neighbourhood of $u_k^{(\ell)}$ is the same as the visible neighbourhood of u_k, so $\forall R.D$ truth is inherited from the original path.

Preservation of existentials. Let $\exists R.D$ be a vertical existential generated at $u_k^{(\ell)}$.

- If $\exists R.D$ was fulfilled within $[u_k, u_j]$ in the original path, the same fulfilling position appears within the same lap as $u_{k'}^{(\ell)}$, $k \leq k' \leq j$.
- Otherwise $\exists R.D \in \mathsf{Pend}(u_j) = \mathsf{Pend}(u_i)$, so the next lap $\ell + 1$ begins with $\exists R.D$ in its pending set, and by request-closure of the segment it is fulfilled inside $[u_i^{(\ell+1)}, u_j^{(\ell+1)}]$.

In both subcases the existential at $u_k^{(\ell)}$ has a strict later R-successor along the pumped path.

Thus every closure formula at every fresh occurrence of the pumped path holds, completing the proof. □

Lemma 8.6 (Finite regularization lemma)**.** *Every generated split presentation satisfying all closure formulas and support-closed mosaic conditions has a finite regular split-forest certificate satisfying Definition 6.4.*

Proof. Label every boundary occurrence by its complete profile. There are finitely many labels. For finite branches, keep one representative context for each complete profile encountered before termination. For an infinite vertical ray, apply Lemma 8.5 and replace the tail by the request-closed repeated segment. This produces a finite stem and a finite blocking cycle. Because repeated laps are fresh, no semantic equality is introduced by this operation.

For branching, perform the same replacement independently at each boundary profile. The substitution lemma guarantees that replacing a component by another component with the same complete profile preserves all interactions through the boundary. Since each profile has only finitely many selected existential obligations and side contexts, only finitely many representative outgoing contexts are needed. Collect these representatives into the finite state set S, record the parent map and child multisets, and record the exact context and mosaic schemes witnessed by the representatives.

Equality data are recorded only as explicit port links inherited from equality of split copies in the original presentation. If two occurrences are merely blocked copies of the same profile, no equality link is recorded. Since the original presentation quotients to a strong frame, no explicit equality link connects strict vertical comparables, and equality mates have identical types and identical external relation behaviour. Hence the typed equality congruence clauses hold.

The reachability summaries in the profiles are copied from the actual representative components and then recomputed on the finite quotient. Request-closed cycles guarantee that recomputation agrees with the copied summaries: a target available after another lap is represented by reachability around the blocking cycle, while distinct laps remain fresh occurrences. Side witnesses and triple support are preserved by the collected support-closed mosaic family. Therefore all validity conditions (V1)–(V10) hold. □

Theorem 8.7 (Completeness of certificates)**.** *If C_0 is satisfiable, then C_0 has a valid finite generated split-forest certificate.*

Proof. Let $\mathcal{I}, a_0 \models C_0$. By Lemma 3.7, replace $\mathcal{I}$ by a witness-generated induced submodel $\mathcal{J}, a_0$ preserving all closure formulas. By Lemma 4.3, construct a generated split presentation of $\mathcal{J}$. This presentation satisfies all closure formulas by pullback from $\mathcal{J}$, and its finite side interfaces are legal because their labels are inherited from the strong RCC5 frame $\mathcal{J}$.

Apply the finite regularization lemma to obtain a finite certificate. All validity clauses are inherited from the presentation and preserved by substitution. The distinguished profile contains C_0 because the distinguished occurrence has origin a_0 and $a_0 \models C_0$. □

9 Decision procedure

This section makes the decision procedure effective by giving (i) a compact formal grammar for certificates, (ii) explicit polynomial bounds on each component alphabet, and (iii) an enumeration bound sufficient to make the satisfiability test an exhaustive finite search. The bound is coarse; computability is what is needed for decidability.

9.1 Grammar of certificate components

Fix the input C_0 and set $n := |\mathsf{cl}(C_0)| \leq 2|\mathrm{sub}(C_0)|$. A finite generated split-forest certificate is the formal expression of the following abstract grammar (writing $\sqcup$ for choice and $\langle \cdot, \dots, \cdot \rangle$ for tuples):

$$
\begin{aligned}
\text{Certificate } \mathcal{Q} &::= \langle S, s_0, \lambda, \mathsf{par}_Q, \mathsf{Ch}_Q, \mathsf{Ctx}_Q, \mathsf{Mos}_Q, E_Q, \mathsf{wit}_Q \rangle \\
\text{StateSet } S &::= \emptyset \sqcup \{\sigma\} \cup S, \quad \sigma \in \Sigma \\
\text{Profile } \sigma &::= \langle \tau, \kappa, \mu, \eta \rangle \\
\text{Type } \tau &\subseteq \mathsf{cl}(C_0), \quad \tau \in \mathsf{Typ}(C_0) \\
\text{ContextColour } \kappa &::= \langle \kappa_\uparrow, \kappa_\downarrow, \kappa_{\text{side}} \rangle \\
\text{SideMenu } \mu &::= \{\, \exists R.D \mapsto (M_i, q_i) : \exists R.D \in \tau,\ R \in \{\mathsf{DR}, \mathsf{PO}\} \,\} \\
\text{InterfaceSummary } \eta &::= \{\, (\pi, R) \mapsto \mathsf{Reach}_R\,(\pi) \subseteq \mathsf{cl}(C_0) : \pi \in \mathsf{Port}(\sigma),\ R \in \{\mathsf{PP}, \mathsf{PPI}\} \,\} \\
\text{ParentMap } \mathsf{par}_Q &::= S \rightharpoonup S \\
\text{ChildMultiset } \mathsf{Ch}_Q(s) &\subseteq_{\text{mset}} S \\
\text{Context } \mathsf{Ctx}_Q &::= \{\, s \mapsto \text{context scheme} \,\} \\
\text{Mosaic } M &::= \langle P, \tau_M, L_M \rangle \\
\text{PositionSet } P &\subseteq \{1, \dots, m(n)\} \\
\text{Edge } L_M(p, q) &\in \mathsf{B} \\
\text{EqPort } E_Q &::= \{(\sigma_i, p_i) \sim (\sigma_j, p_j)\} \\
\text{Witness } \mathsf{wit}_Q &::= \{\, \exists R.D \mapsto \text{mechanism} \,\}
\end{aligned}
$$

Each terminal ranges over a finite alphabet determined by n:

- $\mathsf{cl}(C_0)$: $\leq 2|\mathrm{sub}(C_0)| \leq 2n$.
- $\mathsf{Typ}(C_0)$: $\leq 2^n$.

- $\mathsf{Port}(\sigma)$: bounded by $n+1$ (one port per PP/PPI obligation, plus the identity port).
- Context colours κ: three polynomial slots, each carrying one of finitely many context scheme names.
- Mos_Q: mosaics with position count $\leq m(n)$ where $m(n)$ is the crude width fixed below (one source, one position per existential formula in τ, one equality-mate boundary position per containment witness, and one position per realised triple of such positions).

9.2 Explicit component bounds

Fix a polynomial bound $m(n) = c \cdot n^k$ for the mosaic position count, with constants c, k determined by the context schemes (the choice $c = 8$, $k = 2$ already suffices: each existential $\exists R.D \in \tau$ contributes one source and one witness position, and triple closure quadruples this). Then:

- **Closure types.** $|\mathsf{Typ}(C_0)| \leq 2^n$.
- **Equality-mate interface ports.** $|\mathsf{Port}(\sigma)| \leq n+1$, so the port alphabet across profiles has size $\leq |\Sigma| \cdot (n+1)$.
- **Selected-parent and selected-child slots.** par_Q is a partial function $S \to S$, giving $|S|^{|S|+1}$ choices; Ch_Q is a multiset on S with capacity bounded by the existentials in $\lambda(s)$, so $|\mathsf{Ch}_Q(s)| \leq n$ and the number of child assignments per state is $\leq n \cdot |S|^n$.
- **Request summaries.** η records, per port and per vertical direction, a subset of $\mathsf{cl}(C_0)$: $(n+1) \cdot 2 \cdot 2^{2n} = O(n \cdot 2^{2n})$ bits per profile.
- **Sibling/support mosaics.** A mosaic is a function $\tau_M : P \to \mathsf{Typ}(C_0)$ together with $L_M : P^2 \to \mathsf{B}$: $\leq (2^n)^{m(n)} \cdot 5^{m(n)^2} \leq 2^{p_1(n)}$ for $p_1(n) := n \cdot m(n) + m(n)^2 \cdot \log_2 5$. Mos_Q is a subset: $\leq 2^{2^{p_1(n)}}$ families. Maximum arity is $m(n)$.
- **Finite graph size after blocking.** By Lemma 8.6 the state set S has $|S| \leq N(C_0)$, the number of complete boundary profiles, and $N(C_0) \leq |\mathsf{Typ}(C_0)| \cdot (\text{context colour count}) \cdot (\text{side menu count}) \cdot (\text{interface count})$. Each factor is at most $2^{p_2(n)}$ for an explicit polynomial p_2, so $|\Sigma| \leq 2^{p(n)}$ and $|S| \leq 2^{p(n)}$.

Collecting these,

$$p(n) := n + 2 \cdot m(n) \cdot \log_2 5 + 2n \cdot (n+1) + \text{const}$$

is a polynomial of degree $\max(k+1, 2k)$ in n.

9.3 Enumeration bound

Lemma 9.1 (Certificate enumeration bound)**.** *There is an explicit polynomial $p(n)$ such that the number of finite generated split-forest certificates for C_0 with state count $\leq B(C_0) := 2^{2^{p(n)}}$ and context width $\leq B(C_0)$ is at most $2^{2^{q(n)}}$ for some polynomial $q(n)$. Each certificate's validity (V1)–(V10) is decidable in time polynomial in its description size.*

Proof. By the grammar of §9.1, a certificate is a finite tuple over alphabets of size $\leq 2^{p(n)}$ (states, types, contexts, mosaic positions, port symbols, witness mechanisms) with at most $B(C_0) = 2^{2^{p(n)}}$ components. The total number of certificates of size $\leq B(C_0)$ is then bounded by

$$\left(2^{p(n)}\right)^{B(C_0)} \cdot (\text{combinatorial overhead}) \;\leq\; 2^{p(n) \cdot 2^{2^{p(n)}}} \;\leq\; 2^{2^{q(n)}}$$

for some polynomial q.

Each validity clause (V1)–(V10) is a finite-quantifier predicate over the certificate's components: (V1)–(V3) are local type/context checks; (V4)–(V5) are graph-reachability checks on S (linear in $|S|$ per query); (V6)–(V8) are mosaic-position membership and triple checks (polynomial in mosaic count); (V9) is a congruence check on the explicit-port relation; (V10) is exact summary computation by reachability. All checks run in time polynomial in $|\mathcal{Q}|$, which is polynomial in $B(C_0)$. □

The compressed label `lem:bound` is retained as a synonym of Lemma 9.1 for back-reference compatibility.

Lemma 9.2 (Effective finite search bound; alias of Lemma 9.1)**.** *For each input C_0 there is a computable number $B(C_0)$ such that if C_0 has a valid finite generated split-forest certificate, then it has one with at most $B(C_0)$ states and with context width at most $B(C_0)$.*

9.4 Decidability

Theorem 9.3 (Decidability)**.** *Concept satisfiability for $\mathcal{ALCI}_{\mathrm{RCC5}}$ over strong abstract RCC5 frames is decidable.*

Proof. Given C_0, compute $\mathsf{cl}(C_0)$ and the bound $B(C_0)$. Enumerate all finite tuples of the form in Definition 6.3 with at most $B(C_0)$ profile states and context width at most $B(C_0)$. By Lemma 9.1 this enumeration is finite and each tuple's validity (V1)–(V10) is decidable in polynomial time.

If a valid certificate is found, accept; by Theorem 7.7 (soundness), C_0 is satisfiable. If no valid certificate exists below the bound, reject; by Theorem 8.7 (completeness) and Lemma 9.2, every satisfiable C_0 would have appeared in the finite search. Hence the algorithm halts with the correct answer. □

10 How the repaired proof handles the known stress cases

10.1 Infinite upward chains

The formula

$$C_{\uparrow} = \exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$$

is represented by a certificate with a blocking parent self-loop on a profile containing $C_{\uparrow}$. The unfolding is

$$u_0 \mathsf{PP} u_1 \mathsf{PP} u_2 \mathsf{PP} \cdots,$$

where each u_i is fresh. The existential $\exists \mathsf{PP}.\top$ at u_i is witnessed by u_{i+1}. No semantic equality is introduced by the self-loop.

10.2 Several incomparable upward witnesses

Consider a node requiring two incompatible superpart witnesses:

$$\exists \mathsf{PP}.A \sqcap \exists \mathsf{PP}.B$$

together with restrictions forcing the A- and B-witnesses to be incomparable. A single occurrence has only one parent chain, so the split forest creates two equality-mate occurrences of the same semantic element. One mate is placed below an A-superpart and one below a B-superpart. In the quotient these mates become one element. Existential discharge is equality-aware: an occurrence may satisfy its $\exists\mathsf{PP}.A$ obligation through the parent chain of one mate and its $\exists\mathsf{PP}.B$ obligation through the parent chain of another mate. Typed equality congruence ensures that all universal restrictions and external RCC5 labels remain invariant under the collapse.

10.3 DR/PO witnesses

A DR or PO existential is never discharged by a raw pair table entry detached from a concrete occurrence. It is discharged only by a named side position in a legal mosaic containing the source, the witness, and the relevant boundary positions. Thus relation support and witness support are joint data, and the proof does not combine witnesses from incompatible contexts.

11 What remains of the original split-forest architecture

The repaired proof keeps the original architecture in the following precise form.

Original component	Repaired status
Tree-shaped containment backbone	Survives as a generated witness-cover tree, not as the Hasse diagram of an arbitrary starting model.
Split copies	Survive; they are essential for elements with several selected containment parents.
Sibling interfaces	Survive as support-closed mosaics rather than raw pair tables.
DR/PO sibling simplification	Survives for ordinary non-core sibling frontier pairs after containment, core copies, and equality interfaces are accounted for.
Typed EQ quotient	Survives as an explicit congruence; it is never inferred from profile equality or blocking.
Finite quotient	Survives as a finite regular certificate whose unfolding may be infinite.
Blocking	Survives as finite-index repetition with fresh laps and request-closed cycles.

The proof therefore does not abandon the split-forest idea. It makes explicit the two preliminary reductions needed to reach the intended model class: witness thinning and generated cover presentation. Once those are in place, the proof operates on the sparse generated structures originally envisioned, while the finite certificate records enough equality, reachability, and mosaic information to make the construction sound.